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Exam questions

Question 1: The value of a real estate investment is fundamentally an assessment of its risks and opportunities.

a- Discuss the most relevant risks to bear in mind when investing in real estate. Which of these risks are typical of real estate? Which instead are common to other financial investments? (10 points)

These issues were extensively discussed in Lecture 1 and then again in Lecture 3.

b- What are the main sources of opportunities attached to a real estate investment? How can 'good management' add significant value to your investment? Discuss using insights from both the lectures and the case study interventions. (15 points)

These issues were also extensively discussed in Lecture 1.

c- 'Inter-lease and intra-lease discount factors can be applied to different parts of a cash flow to take into account their different risks'. Discuss and explain this statement (10 points)

These issues were extensively discussed in Lecture 1.

d- Consider the following investment. For the first four years, you have already signed up a tenant paying rents worth 1 million in years 1 and 2, and worth 2 million in years 3 and 4. You are planning to keep the investment till the end of period 6 and expect to sell it for 20 million. You are also expecting to earn 3 million in rents in periods 5 and 6, but you have not signed any tenant up yet. Using a discount factor of 10%, how much is this investment worth? (15 points)

This is a simple application of a discounted cash flow. The table below represents the stream of payments

	T=0	T=1	T=2	T=3	T=4	T=5	T=6
NOI		1	1	2	2	3	3 + 20
Sale/Purchase price	?						Note: include sale value

Using a DCF, the students should find a value of 19.45 (approximately)

e- How would your answer change if you applied an 'inter-lease' discount factor of 12% and an 'intra-lease' discount factor of 7%? Explain the rationale behind your calculations and discuss your findings (15 points)

The students should apply the following formula and find that the purchase price has now decreased to 18.55 because the investment is considered more risky. I expect students to clarify why they are applying different rates to different part of the cash flow to reflect tenancy/lease risks.

$$PV = \frac{1}{(1+7\%)} + \frac{1}{(1+7\%)^2} + \frac{2}{(1+7\%)^3} + \frac{2}{(1+7\%)^4} + \frac{1}{(1+12\%)^4}\left[\frac{3}{(1+7\%)} + \frac{3}{(1+7\%)^2}\right] + \frac{20}{(1+12\%)^6}$$

f- Using the price you obtain at point e-, calculate the Internal Rate of Return (IRR) of this investment. Discuss how it compares to the inter-lease and intra-lease discount factors. (5 points)

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The IRR is 11.02% which sits (expectedly) between the inter-lease and intra-lease discount factors.

g- Assume you used the intra-lease discount factor (7%) as a going-in cap rate. What price would you be willing to pay? What problems do you see with the answer from this calculation and with the application of a simple cap-rate method to this example? (10 points)

The price would be 14.286 which is clearly lower than at the previous point because cap-rate does not incorporate any information on the growth in the value of the rents (and resale price).

h- In UK practice, yields are used to provide a first-pass valuation of possible real estate investments. Discuss the yields method using some of the concepts presented by the first case-study lecturer (including term-and-reversion; layer-and-hard core; equivalent yield; net initial yield; and reversionary yield). (20 points)

This topic was discussed by the lecturer during Lecture 2 and then again by the first case-study practitioner in week 4 of the course.

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Question 2: Investors can benefit from the stable returns of real estate while avoiding its risks by investing in real estate *indirectly*.

a- What are the main risks that investors purchasing shares of Real Estate Investment Trusts (REITS) or Mortgage-Backed Securities (MBS) are trying to avoid? And in which ways do these *indirect* investments – at least in theory – by-pass these risks? (10 points)

The risks are liquidity and lack of information. These topics were discussed in Lecture 1 and then again on the specific lectures on REITS (Lecture 6) and on MBS (Lecture 7 and following).

b- What are the main constraints that REITs face in terms of investment? How can they 'add value' given these constraints? Discuss using the insights put forward by your regular lecturer as well as the guest speaker for the lecture on REITs. (20 points)

These were the core issues of Lecture 6 on REITs.

c- The process of mortgage securitization can be initiated via either a 'cash operation' or a 'swap programme'. How do these two approaches work, and how do they allocate the risks of the underlying mortgage portfolios to different economic agents in the chain of transactions? (15 points)

These issues were discussed at the beginning of Lecture 7 and then again in lecture 8 on MBS and sub-prime mortgages.

d- Discuss the main features of Collateralized Mortgage Obligations (CMOs) and Mortgage Pass-Through Securities, highlighting similarities and differences in terms of allocation of risk to investors. How does tranching work? In what ways does tranching affect the risk that different investors face? Provide some simple examples to clarify your discussion. (25 points)

These issues were discussed in Lectures 7 and 8. I expect students to discuss in some details overcollateralization, tranching and the different types of income/risk passed to investors.

e- Some of the benefits of securitization did not materialise and a global financial crisis emerged from troubles in securities backed by sub-prime mortgages. What are subprime mortgages and in what sense is the profitability and risk of this type of loan intrinsically linked to the dynamics of house prices? (15 points)

This was the core of the discussion of the Lecture 8 and Lecture 9.

f- Why are securities backed by sub-prime mortgages particularly risky and complicated to manage? How did these features contribute to the sub-prime mortgage crisis of 2007? (15 points)

This was also a topic discussed during Lecture 8 and concluded in Lecture 9.

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Question 3: You are planning to buy a property worth £1,500,000 and the bank is offering a loan with an LTV of 75%. The loan has monthly compounding and stretches over 30 years. The annual interest rate is 4.75%.

a- The bank is initially offering you to choose between an Interest Only (IO) mortgage, a Constant Amortization Mortgage (CAM) and a Constant Payment Mortgage (CPM). What are the payments for these three alternative mortgages in the first, the second, the 120th and 359th month? (15 points)

These types of calculations were discussed several times during the course. The first thing to realise is that the value of the loan is 75% of £1,500,000, that is £1,125,000.

For the IO, the interest payments are 4.75%/12*£1,125,000 = 4453.12. This is the only cost in months 1, 2, 120 and 359. Students might want to add that the spike reimbursement payment happens in period 360.

For the CPM, the payments are constant at 5868.53.

The CAM is slightly more complicated. First, note that the constant amortization is 1125000/360=3125.

Here is the schedule of payments for the different months:

1st: 3125 + 4.75% / 12 * (1125000) = 7578.12

2nd: 3125 + 4.75% / 12 *(1125000 – 3125) = 7565.75

120th: 3125 + 4.75% / 12 * (1125000 – 119*3125) = 6106.12

359th: 3125 + 4.75% / 12 * (1125000 – 358*3125) = 3149.74

b- Consider the CPM mortgage. Provide the following quantities for the 1st, 2nd, 120th and 359th periods: total amount of interests paid up to and including that period; total amount of amortization contributions made up to that period; outstanding loan balance at that moment. (10 points)

These quantities can be easily worked out with the calculator. They are presented in the table below:

Month	Total interest	Total amort.	Outstanding balance
1	4453.12	1415.41	1123584.59
2	8900.65	2836.42	1122163.58
120	487351.11	216872.79	908127.206
359	987648.59	119154.58	5845.42

c- Using the standard formula that characterise the payment, amortization and outstanding balance of a loan, show how you would manually calculate the total interest paid, the total amortization paid and the outstanding balance on the CPM loan at the end of the first and second months. Do these numbers square up with those provided at point b-? (15 points)

These numbers can be obtained as follows:

PMT=5868.53. In period 1, interest is equal to 1125000*4.75%/12=4453.12. PMT-INT=AMORT=1415.41. The BAL1=BAL0-AMOR=1125000-1415.41=1123584.59. The INT on this is 4447.52. Since PMT stays constant, we can use PMT-INT2=AMORT2=1421.01. Total amort and total interest can be obtained from summing first and second month numbers. The BAL2=BAL1-AMOR=1123584.59-1421.01=1122163.58.

d- Your monthly income is £7,200. If the bank uses a minimum debt-coverage-ratio (DCR) of 1.2, which loan do you qualify for? (5 points)

The person qualifies for the IO and CPM loan only.

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e- You are now attempting to reduce the monthly payments on the CPM loan to be closer to the monthly payments on the IO loan. If you reduce the CPM monthly payments to £5000, what is the outstanding loan due at the end of the contractual period (i.e. after 30 years)? And what if this number if you further reduce to £4453.12? What do you notice? What would instead be the payments if you wanted to have an outstanding loan at the end of the contractual period of £300,000? (20 points)

These are very simple applications of partially amortizing loans where either FV is the unknown or FV is given but not equal to zero. The answer to the first question is: 690330.58. The answer to the second question is approximately 1125000. The students should note that this is the full value of the loan. This is not surprising because they have reduced the CPM to the IO payments and so they do not contribute to amortizing the loan. The answer to the last question is 5491.09.

f- You have decided to consider the IO and CPM loans only. The bank is applying two discount points. What is the effective cost of borrowing for the CPM loan if you hold the loan till maturity? And what is the effective cost of borrowing for the IO loan if hold it to maturity? How do these numbers change if you repay after 10 years? Comment on your results. (20 points)

The effective amount of money borrowed is 1102500 even though both the CPM and the IO payments stay the same.

The IRR on the CPM can be easily calculated using the time-value worksheet and the amortization worksheet of calculator. If the person holds to maturity, the IRR is 4.9258%. If the person is repaying after 120 periods, the IRR is 5.0270%.

For the IO loan, the students will have to use the cash-flow worksheet and consider the following. If they hold to maturity, they have a PV= 1102500, then they have 359 periods of 4453.12 and finally one period of 4453.12 + the full loan 1125000 (not the value with discount points – but the full loan). If they repay after 120 periods, the expressions have to be changed accordingly. The two IRR are 4.877% and 5.005%.

The IRRs increase in all cases as we repay early. This is expected and the students should explain why (a sunk cost is reimbursed over a shorter period of time). However, the IRR of the CPM increase more because included in the payments is a contribution towards amortization (which instead is deferred in the IO loan).

g- You are now offered an adjustable rate mortgage. The structure is as follows. The loan lasts 30 years and has monthly compounding. The initial interest rate is 3.5%. This is expected to climb to 4.5% from the beginning of the third year to the end of the fourth, and to further increase to 5% from the beginning of the 5th year till the end of the contract. The bank is applying 2 discount points. What are the payments you owe to the bank in in month 1, month 25 and month 49? And what is the IRR on this loan if you hold it to maturity? (15 points)

ARMs were discussed at length during the lectures. They can be treated as difference CPM recalculated on the outstanding loan whenever interests adjust.

For the first 24 months, the payment is 5051.753. At the end of these 24 months, this gives an outstanding loan of 1081051.72. This can be used to calculate the next tranche of payments with an interest of 4.5%. This gives 5664.49. At the end of the 48th period the balance is 1040684.93. With an interest of 5%, this gives 5966.70. The overall IRR with discount points is 4.858%.

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Question 4: You are considering investing in the following building. The overall space consists of 25,000 square metres, 15,000 on the ground floor and 10,000 on the first floor. Your holding period is five years. You then expect to sell the asset. The top floor is office space and is occupied by a company currently paying £180 per square metre per year. They have signed a contract for the first two years of your investment period. After that, there is a 50% chance that they will stay and continue to pay the same rent. But with 50% probability they will leave. If this happens, you will have to refurbish the place. This will cost £25 per square metre and works will take 6 months. During these 6 months you won't be able to let the floor. After that, you expect to earn £210 per square metre. No concessions will be given to tenants. On the ground floor, you are hoping to have three tenants. The first one, taking 10,000 meters, is a large retailer that will pay £200 per square metre. However, you are planning to give one and a half year free-rent period to this tenant. If the tenant is occupying the space, you expect to generate additional revenues through the parking lot. You expect these to come from 200 cars using the parking lot each of the 300 working days per year when the retailer is open. Each car is charged £2 per day. You are hoping to let the other 5,000 square metres to two smaller shops – each occupying 2,500 square metres. They will be paying £200 per square metre and will benefit from one rent-free year. After that, they will also be charged an overage rent at 10% of their revenues. These are expected to be £1,000,000 per year. Your expected exit yield is 8%.

a- Put together the operation and reversion pro-forma for this investment. (20 points)

These issues were discussed in several lectures. The exit yield is the same concept as the going out cap rate. The pro-forma for the operations looks as follows:

	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Tenant 1	0	50% of 10000m2 × 200= 1,000,000	10000 m2 × £200 = £2,000,000	10000 m2 × £200 = £2,000,000	10000 m2 × £200 = £2,000,000	10000 m2 × £200 = £2,000,000
Parking lot	120,000	120,000	120,000	120,000	120,000	120,000
Tenant 2	0	2500m2 × £200 = £500,000	2500m2 × £200 = £500,000	2500m2 × £200 = £500,000	2500m2 × £200 = £500,000	2500m2 × £200 = £500,000
Overage rent	0	10% of 1,000,000= 100,000	10% of 1,000,000= 100,000	10% of 1,000,000= 100,000	10% of 1,000,000= 100,000	10% of 1,000,000= 100,000
Tenant 3	0	2500m2 × £200 = £500,000	2500m2 × £200 = £500,000	2500m2 × £200 = £500,000	2500m2 × £200 = £500,000	2500m2 × £200 = £500,000
Overage rent	0	10% of 1,000,000= 100,000	10% of 1,000,000= 100,000	10% of 1,000,000= 100,000	10% of 1,000,000= 100,000	10% of 1,000,000= 100,000
Office building – stay	1,800,000	1,800,000	1,800,000 x 50% = 900,000	1,800,000 x 50% = 900,000	1,800,000 x 50% = 900,000	1,800,000 x 50% = 900,000
Office building – leave	n/a	n/a	210 x 10000 m2 x 50% x 6months out of 12 =	210 x 10000 m2 x 50% = 1,050,000	210 x 10000 m2 x 50% = 1,050,000	210 x 10000 m2 x 50 = 1,050,000

			525,000			
Refurb. cost	n/a	n/a	25 x 10000 x 50% = 125,000	0	0	0
Total	1,920,000	4,120,000	4,620,000	5,270,000	5,270,000	5,270,000

With an exit yield of 8%, the sale price is £65,875,000.

b- If you are asked to pay £50million to purchase this building, what is the Internal Rate of Return (IRR) of the investment? (5 points)

The IRR of this investment for a price of £50mio is 12.9871%.

c- You decide to gear your investment and use an interest only (IO) loan with an LTV of 70% and charging an annual interest rate of 5% (consider for simplicity annual payments and annual compounding). You plan to reimburse the loan when you sell the building in year 5. What is the IRR of the investment after leverage? Comment on your finding. (10 points)

The IRR becomes 26.5496%. Gearing has a positive effect on the IRR.

d- What is the return on equity if you use the Weighted Average Cost of Capital? Do you get the same figure you obtained at point c-? Why/why not? (5 points)

The WACC gives approximately 31%. This is larger than the correct IRR but it is not surprising since the WACC is an approximation. In this case, it holds the value of D/E constant at the beginning of time (whereas it is growing because of the appreciation of the equity).

e- What are the cash-on-cash yields in year 1 and year 5? And what is the annualised growth yield of this investment? Comment on your findings discussing whether the patterns of results makes this investment seemingly risky or not. (10 points)

The cash-on-cash yields are as follows:

C-C1= [(1,920,000/50,000,000)-5%*70%]/[1-70%]=1.13%

C-C5= [(5,270,000/50,000,000)-5%*70%]/[1-70%]=23.47%

The growth yield can be obtained by comparing the value of the initial investment (15mio) and the final equity after repaying the loan (30,875,000). In order to get the annualised growth yield, the students have to solve the following equality with r as an unknown:

(15,000,000)*(1+r)^5=30,875,000 → [(30,875,000/15,000,000)^1/5]-1 = 15.53% per annum

The growth yield is very good but the cash-on-cash is clearly dependent on the performance of the retailers and the two smaller shops. Unless they occupy the space, the yield is very low.

f- In order to stress-test your numbers, you consider the following alternative scenario. First, the probability that the tenant occupying the office space on the first floor leaves increases to 70%. All other numbers regarding this part of your space remain the same. Second, you expect the main tenant on the ground floor to leave in year 3. During that year, you do not manage to let the space but the leaving tenant has agreed to pay a compensation of £150 per square metre while the space remains vacant. You manage to let the space again at the beginning of year 4 at the same rent as before. When the space is not occupied, you expect to have no revenues from the parking space. You also expect the revenues for the other two smaller shops to be disrupted

and reduced to £700,000 while the main ground-floor tenant is not in place. Put together the pro-forma for this alternative scenario and calculate the implied IRR with and without leverage. (15 points)

Although this looks very different from the first scenario, many of the numbers remain the same.

	Year 1 (SAME)	Year 2 (SAME)	Year 3	Year 4	Year 5	Year 6
Tenant 1	0	50% of 10000m2 × 200= 1,000,000	10000 m2 × £150 = £1,500,000 (compensation)	10000 m2 × £200 = £2,000,000	10000 m2 × £200 = £2,000,000	10000 m2 × £200 = £2,000,000
Parking lot	120,000	120,000		120,000	120,000	120,000
Tenant 2	0	2500m2 × £200 = £500,000	2500m2 × £200 = £500,000	2500m2 × £200 = £500,000	2500m2 × £200 = £500,000	2500m2 × £200 = £500,000
Overage rent	0	10% of 1,000,000= 100,000	10% of 700,000= 70,000	10% of 1,000,000= 100,000	10% of 1,000,000= 100,000	10% of 1,000,000= 100,000
Tenant 3	0	2500m2 × £200 = £500,000	2500m2 × £200 = £500,000	2500m2 × £200 = £500,000	2500m2 × £200 = £500,000	2500m2 × £200 = £500,000
Overage rent	0	10% of 1,000,000= 100,000	10% of 700,000 = 70,000	10% of 1,000,000= 100,000	10% of 1,000,000= 100,000	10% of 1,000,000= 100,000
Office building – stay	1,800,000	1,800,000	1,800,000 x 30% = 540,000	1,800,000 x 30% = 540,000	1,800,000 x 30% = 540,000	1,800,000 x 30% = 540,000
Office building – leave	n/a	n/a	210 x 10000 m2 x 70% x 6months out of 12 = 735,000	210 x 10000 m2 x 70% = 1,470,000	210 x 10000 m2 x 70% = 1,470,000	210 x 10000 m2 x 70% = 1,470,000
Refurb. cost	n/a	n/a	25 x 10000 x 70% = 175,000	0	0	0
Total	1,920,000	4,120,000	3,740,000	5,330,000	5,330,000	5,330,000

In this case, the sale price becomes 66,625,000. So the unleveraged IRR is 12.9200% while the leveraged IRR is 26.2419%.

g- Considering that the two scenarios described at point a- and f- happen each with 50% probability, calculate the expected IRR, standard deviation of the return and coefficient of variation for the leveraged and unleveraged case. Comment on your findings. (10 points)

These numbers are easy to work out.

Consider the unleveraged case. The expected IRR is 12.9535%. The standard deviation is 0.0335%. So the CoV becomes 0.2586%.

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In the leveraged case, these become: 26.3958%; 0.1539%; and 0.5829%. Clearly leverage increases expected returns but also the risk as proxied by the CoV.

h- Consider now only the scenario depicted in case a-. You want to understand the after tax return of this investment taking into account the following issues. The tax rate on the income and the capital gains is 30%. Mortgage interest costs are tax deductible. Any negative tax amount in a given year can be used as a tax credit to reduce taxation in the following year. Buildings depreciate on a straight line over 30 years and depreciation is tax deductible. You also estimate that the value of your building is 30,000,000 out of the 50,000,000 of its purchase price. What is the after tax IRR in the leveraged and unleveraged case? Discuss your results. (25 points)

This seems like a very complicated set of calculations to carry out but they were well-rehearsed during Lecture 3.

Note the following: the value of the building depreciates by 1mio every year. This is deductible.

Over 5 years, this means the building should be worth 5mio less. So the value of the investment should be 45mio. It is however sold for 65,875,000. This means there are 20,875,000 capital gains giving rise to a 6,262,500 capital gain taxation.

The calculations for the unleveraged case can be summarised as follows:

	Year 1	Year 2	Year 3	Year 4	Year 5
Total	1,920,000	4,120,000	4,620,000	5,270,000	5,270,000
Deprec.	-1mio	-1mio	-1mio	-1mio	-1mio
After Tax	1,644,000	3,184,000	3,534,000	3,989,000	3,989,000

The purchase price is still 50mio. The net sale value is. 59,612,500 This gives an IRR of 9.5118%.

For the leveraged case things are somewhat more complicated. They can be worked out as follows:

	Year 1	Year 2	Year 3	Year 4	Year 5
Total	1,920,000	4,120,000	4,620,000	5,270,000	5,270,000
Deprec.	-1mio	-1mio	-1mio	-1mio	-1mio
Mortgage deduct.	-1.75mio	-1.75mio	-1.75mio	-1.75mio	-1.75mio
Taxes due	+249,000	-411,000	-561,000	-756,000	-756,000
Tax credits	n/a	249,000	n/a	n/a	n/a
Mortgage paym.	-1,750,000	-1,750,000	-1,750,000	-1,750,000	-1,750,000
After tax income	170,000	2,208,000	2,309,000	2,764,000	2,764,000

This gives an IRR of: 20.62%.

Taxation clearly reduces the IRR in both the unleveraged and leveraged case. But when using leverage the reduction is less significant. This is because some of the mortgage costs can be deducted – which makes borrowing cheaper after tax.